Assignment 6: Application Mapping of Sequential Building Blocks

➢ Flip-Flop: 1-Bit Memory Cell in Keypad Debounce Circuits

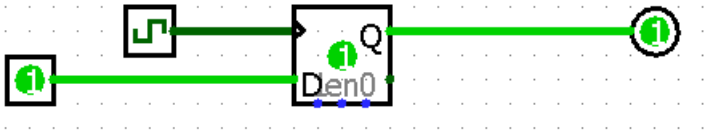
- **Problem Statement:** Combinational circuits cannot retain past data without memory; a basic persistent storage element is required for processor registers.

- **Objective:** To simulate a D flip-flop in Logisim that holds a stable binary state as a foundational register bit

- **Real System:** CPU Register / Microcontroller Bit Storage & Keypad Debounce Circuit

- **Role:** Acts as the fundamental memory element (flip-flop) responsible for storing a single binary digit (0 or 1) persistently until updated by a new clock pulse or cleared.

❖ **Circuit Implementation in Logisim:**



- **Expected Output:** When the input D is set to high (1) and a clock edge is applied, the output Q transitions to high (1) and holds that stable state indefinitely until the input changes or the clock triggers a new value.

- **Observation:** In the Logisim simulation, with input $D = 1$ and an active clock source, output Q remains continuously active (1), as verified by the output indicator probe.

- **Discussion:** This circuit demonstrates the core principle of volatile sequential memory. In real-world microprocessors, thousands of such D flip-flops are combined in parallel to form multi-bit registers (e.g., 8-bit, 32-bit registers) that temporarily hold data during arithmetic and logic operations.

- **Mod-60 Counter / Frequency Divider (Digital Clock Timer)**

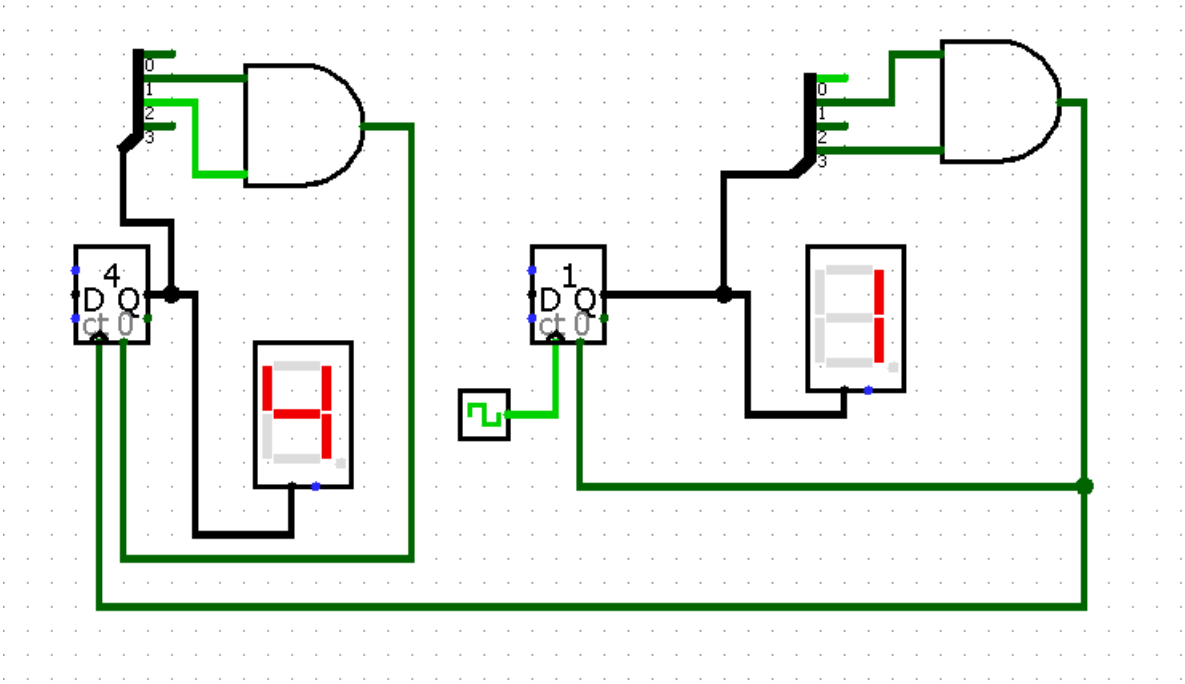
- **Problem Statement:** Standard binary counters count in powers of two, whereas real-world timekeeping requires cyclical counting up to base-60.

- **Objective:** To simulate a Mod-60 counter using sequential blocks and logic gates to track time for a digital clock.

- **Real System:** Digital Clock (Seconds and Minutes Counter) / Frequency Divider

- **Role:** Counts clock pulses sequentially up to a specific modulus (such as base-60 for seconds and minutes in a digital clock) and drives display units to show elapsed time.

❖ **Circuit Implementation in Logisim**



- **Expected Output:** Sequential state transitions representing numerical increments on the 7-segment displays synchronized with the input clock frequency.

- **Observation:** The Logisim circuit successfully cascades flip-flops and logic gates (AND gates/feedback loops) to cycle through digit states, showing active count updates on the 7-segment display modules.

- **Discussion:** Modulus counters are essential in timekeeping systems. By dividing a high-frequency oscillator signal (e.g., a 32.768 kHzquartz crystal) down to 1 Hz, the counter accurately tracks seconds, minutes, and hours, demonstrating the practical application of synchronous sequential state machines.

➢ **Shift Register (SIPO Operation)**

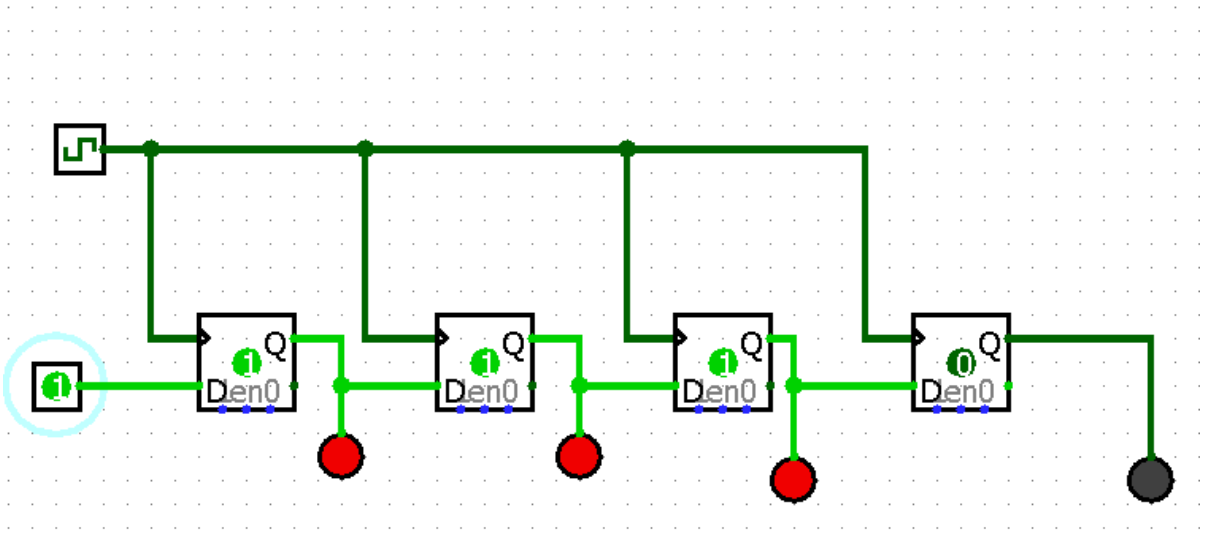
- **Problem Statement:** Multi-bit parallel data transfer requires too many physical wires, necessitating efficient serial-to-parallel conversion.

- **Objective:** To simulate cascaded D flip-flops demonstrating bit-shifting overclock cycles for serial communication.

  Real System: Serial Communication Interface (UART / SPI) & LED Shift Light / Display Driver

- **Role:** Transfers and shifts data bit-by-bit across multiple clock cycles, enabling serial-to-parallel or parallel-to-serial data conversion.

❖ **Circuit Implementation in Logisim:**



- **Expected Output:** With a serial data input ($D = 1$) and a shared clock, data propagates sequentially from the first flip-flop stage to subsequent stages on each consecutive clock edge.
- **Observation:** The simulation shows four cascaded D flip-flops where the high state progressively propagates across the stages (Q outputs turning high one by one, indicated by the red LED probes).
- **Discussion:** Shift registers are vital in digital communication protocols like SPI, I^2C, and UART, where data is transmitted over a single wire line sequentially and then converted into parallel bytes inside a receiver chip, minimizing the number of physical interconnect pins required.